

Devriş İşler

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🔗 Google Scholar

Research Interest. My research has focused on developing trustworthy and verifiable AI and data-driven systems through applied cryptography, secure systems, and privacy-enhancing technologies. Throughout my PhD, I investigated how provenance, ownership, accountability, authenticity, and privacy can be established and independently verified in modern data and machine-learning ecosystems. I am interested in trustworthy AI, including auditing, accountability, and governance. With the advancement of AI in recent years, I believe there is a growing need for mechanisms that support reliable monitoring, verification, and oversight of AI systems. While my expertise lies in applied cryptography and secure systems design, I value interdisciplinary research that connects technical guarantees with governance, human-centered design, and policy. My long-term goal is to contribute to the development of safe, transparent, privacy-preserving, and accountable AI systems, while helping bridge the gap between technical solutions, real-world deployment, and regulatory frameworks.

Education

- Dec. 2025 📖 **Ph.D., Data Transparency Group, IMDEA Networks Institute, Spain.**
Ph.D., Telematics Engineering, Universidad Carlos III de Madrid, Spain
Thesis title: *Data Ownership in the Human-Centric Data Economy Era.*
Supervisor: Prof. Nikolaos Laoutaris.
- 2018 📖 **M.Sc., Cryptography, Security, and Privacy Research Group, Computer Science and Engineering, Koç University, Türkiye.**
Thesis title: *Distributed Single Password Protocols.*
Supervisor: Prof. Alptekin Küpçü.
- 2015 📖 **B.Sc. Computer Science and Engineering, Gaziantep Zirve University, Türkiye.**
- 2013 📖 **[ERASMUS Exchange] B.Sc. Computer Science and Management, Wrocław University of Technology, Poland.**

Research and Work Experience

- 2025-Dec. 2026 📖 **Postdoctoral Researcher, IMDEA Networks Institute, Spain.**
- 2022 📖 **Research Visit (Internship), University of California, Irvine, USA.**
 - 1. Developed Puppy, a framework for publicly verifiable ownership and provenance claims that enables owner-free verification without requiring trusted third party. Puppy explores how cryptographic evidence can support accountability in data and AI ecosystems.
 - 2. Developed FedPoP, a privacy-preserving accountable federated machine learning that enables verifiable participation without compromising participant privacy and anonymity.
- 2019 📖 **Research Assistant, KU Leuven, Belgium.**
Worked on secure multiparty computation.
- 2017 📖 **Short Research Visit, University College London (UCL), UK.**
Visited Prof. Emiliano De Cristofaro at UCL for research on usable and secure password authentication.
- 2015 📖 **Undergraduate Research Assistant, Gaziantep Zirve University, Türkiye.**
Worked in 1001 TUBITAK (Scientific and Technological Research Council of Turkey) Project called as “Secure Computation of the k -th Ranked Element over Shared Data and its Applications” under the supervision of Prof. Mehmet Ercan Nergiz and Prof. Serdar Pehlivanoglu.
- 2014 📖 **Summer Undergraduate Internship, Concordia University, Canada.**
Under the Mitacs Globalink Summer Research Internship Program, worked as an intern under the supervision of Prof. Mohammad Mannan.

Research Publications

Published

- 1 Javad Dogani, **Devriş İşler**, and Nikolaos Laoutaris, “INDEXGUARD: Index-only backdoor vetting for secure federated peft of large language models,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
- 2 Kaan Altınay, **Devriş İşler**, and Zekeriya Erkin, “Extending null embedding for deep neural network watermarking,” in *Proceedings of the 22nd International Conference on Security and Cryptography - SECRYPT*, 2025.  DOI: 10.5220/0013641200003979.
- 3 Mike P. Raave, **Devriş İşler**, and Zekeriya Erkin, “Dataset watermarking using the discrete wavelet transform,” in *Proceedings of the 22nd International Conference on Security and Cryptography - SECRYPT*, 2025.  DOI: 10.5220/0013556300003979.
- 4 **Devriş İşler**, Elisa Cabana, Alvaro Garcia-Recuero, Georgia Koutrika, and Nikolaos Laoutaris, “FreqWM: Frequency watermarking for the new data economy,” in *IEEE International Conference on Data Engineering 2024 (ICDE)*, 2024.  URL: <https://ieeexplore.ieee.org/abstract/document/10597840>.
- 5 **Devriş İşler** and Johanna Gunawan, “Privacy perceptions and behaviors of LGBTQ+ community in Türkiye,” in *Workshop on Technology and Consumer Protection (ConPro’24), co-located with IEEE Symposium on Security and Privacy*, 2024.  URL: <https://conpro24.ieee-security.org/papers/isler-conpro24.pdf>.
- 6 Tianyue Chu, **Devriş İşler**, and Nikolaos Laoutaris, “Strengthening privacy in robust federated learning through secure aggregation,” in *NDSS Workshop on AI System with Confidential Computing, AISCC*, 2024.  URL: <https://www.ndss-symposium.org/wp-content/uploads/aiscc2024-12-paper.pdf>.
- 7 Palle Maesen, **Devriş İşler**, Nikolaos Laoutaris, and Zekeriya Erkin, “Image watermarking for machine learning datasets,” in *Proceedings of the Second ACM Data Economy Workshop, DEC 2023*, ACM, 2023.  DOI: 10.1145/3600046.3600048.
- 8 Tsvetomir Hristov, **Devriş İşler**, Nikolaos Laoutaris, and Zekeriya Erkin, “Graph database watermarking using pseudo-nodes,” in *Proceedings of the Second ACM Data Economy Workshop, DEC 2023*, ACM, 2023.  DOI: 10.1145/3600046.3600049.
- 9 **Devriş İşler**, Alptekin Küpçü, and Aykut Coskun, “User perceptions of security and usability of mobile-based single password authentication and two-factor authentication,” in *Data Privacy Management, Cryptocurrencies and Blockchain Technology - ESORICS 2019 International Workshops, DPM*, 2019.  DOI: 10.1007/978-3-030-31500-9.
- 10 **Devriş İşler** and Alptekin Küpçü, “Threshold single password authentication,” in *Data Privacy Management, Cryptocurrencies and Blockchain Technology - ESORICS 2017 International Workshops, DPM*, 2017.  DOI: 10.1007/978-3-319-67816-0.

Under Submission

- 1 **Devriş İşler**, H. S. Dadmarzi, and A. Küpçü, “Secure and updatable single password authentication,” in 2026.  URL: <https://eprint.iacr.org/2026/784>.
- 2 **Devriş İşler**, Elina van Kempen, Seoyeon Hwang, and Nikolaos Laoutaris, “FedPoP: Federated learning meets proof of participation,” in *Under Submission*, 2025.  URL: <https://arxiv.org/pdf/2511.08207>.
- 3 Javad Dogani, Mohammad Mojtaba Roshani, **Devriş İşler**, Tianyue Chu, and Nikolaos Laoutaris, “SAFE-PFL: Efficient and secure personalized federated learning,” in *Under Submission*, 2025.
- 4 **Devriş İşler**, Seoyeon Hwang, Yoshimichi Nakatsuka, Nikolaos Laoutaris, and Gene Tsudik, “Puppy: A publicly verifiable watermarking protocol,” in *Under Submission*, 2023.  URL: <https://arxiv.org/pdf/2312.09125>.

Projects

- 2022-2025  **DataBri-X Project** (<https://databri-x.eu>)
Developed and integrated FreqyWM [4] into the DataBri-X platform, contributed to project deliverables, and co-organized two project datathons.
- 2023-2025  **COST Action CA20137 – Making Young Researchers’ Voices Heard for Gender Equality (VOICES)**
Participated in this project addressing gender and intersectional inequalities affecting young researchers and innovators, contributing to international collaboration and exchange on equality, inclusion, and research policy in academic environments.
- 2021  **DW-Marking, Ontochain Project** (<https://networks.imdea.org/projects/dw-marking/>)
Contributed to the project proposal and technical deliverables and investigated NFTs and their potential role in proof of dataset ownership in data economy settings.

Supervision and Mentoring

- B.Sc. Projects  **TU Delft**, the Netherlands [**Main advisor:** Prof. Zekeriya Erkin] : **1)** Kaan Altınay; **2)** Matthijs C. van Anel; **3)** Tsvetomir Hristov; **4)** Palle Maesen; **5)** Mike P. Raave; **6)** B. Benedek Székács; **7)** Ruonan Li; **8)** Craciun Marius-Cosmin; and **9)** Jaden Nierop.
-  **Koç University**, Türkiye [**Main advisor:** Prof. Alptekin Küpçü]: **1)** Arjen Kılıç; and **2)** Ilker Kadir Ozturk.

Selected Invited Talks and Presentations

- 2025  ***A Bermuda Triangle? Data Ownership, Data Privacy, and Data Economy.*** Presented at **1)** The Twenty-First Symposium on Usable Privacy and Security (SOUPS Lighting Talk 2025 as a short version), Seattle, USA; **2)** Istanbul Bilgi University, Türkiye; and **3)** Koç University, Türkiye.
- 2024  ***Students’ Perception of LGBTQ+ : How (Not) to Conduct a Study with At-Risk Populations from a Queer Lens.*** Presented at the VOICES Seminar (<https://gendervoices.eu>).
-  ***FedPoP: Federated Learning Meets Proof of Participation.*** Presented at IMDEA Networks Institute.
- 2023-2024  ***FreqyWM: Frequency Watermarking for the New Data Economy.*** Presented at **1)** Athena Research Center, Greece; **2)** International Conference on Data Engineering, the Netherlands (2024); **3)** IMDEA Networks Institute, Spain; and **4)** The ProperData Weekly Seminar, USA (<https://properdata.eng.uci.edu/>).
- 2023  ***Privacy Perceptions and Behaviors of LGBTQ+ Community in Türkiye.*** Presented at **1)** Breaking the Bias: Inspire Diversity in R&I Summer School, Greece (**Best Poster Award**); **2)** the Workshop on Technology and Consumer Protection 2024 (ConPro’24), co-located with IEEE Symposium on Security and Privacy; and **3)** the ProperData Second Annual Symposium, Boston.
- 2022  ***Puppy: A Publicly Verifiable Watermarking Protocol.*** Presented at **1)** University of California, Irvine; and **2)** IMDEA Networks Institute, Spain.

Academic Service and Volunteering

- 2022-2025  **Co-organisier** of Nothing to Hide? A Data Privacy Podcast (<https://nothingtohide.online>).
- 2023-2024  **Ph.D. Representative** at IMDEA Networks Institute.

Academic Service and Volunteering (continued)

- **Board Member** at The European Council of Doctoral Candidate and Junior Researchers (EuroDoc) (<https://www.eurodoc.net>).
- **Co-organizer of In-house Seminars** at IMDEA Networks Institute.
- Past ■ **Program Committee and Reviewer**
 - **Program Committee:** International Conference on Security and Cryptography (SECRYPT 2025), NDSS Workshop on AI System with Confidential Computing (AISCC 2024).
 - **Reviewer:** The VLDB Journal, Journal of Information Security and Applications (JISA), IEEE Transactions on Information Forensics and Security (IEEE TIFS), IEEE Transactions on Big Data (IEEE TBDATA), ACM Conference on Data and Application Security and Privacy (CODASPY), International Conference on Information Security and Cryptology (ICISC), International Conference on Security and Cryptography (SECRYPT), International Conference on Cryptology in India (Indocrypt).

Awards and Grants

- 2024 ■ **Travel Grant and Volunteer**, Internet Measurement Conference (ACM IMC), Spain.
- 2023 ■ **Travel Grant**, Privacy Enhancing Technologies Symposium (PETS), Switzerland.
- **Best Poster Award**, *Privacy Perceptions and Behaviors of LGBTQ+ Community in Türkiye* by Breaking the Bias: Inspire Diversity in R&I Summer School, Greece.
- **Travel Grant**, Breaking the Bias: Inspire Diversity in R&I Summer School, Greece, <https://gendervoices.eu/activities/training-schools/breaking-the-bias-summer-school/>
- **Summer School Stipend**, Summer School on Real-World Crypto and Privacy, Croatia.
- 2022 ■ **Winter School Grant**, the 12th Bar-Ilan Winter School on Cryptography.
- 2021 ■ **ACM CCS Grant**, Supported by iMentor Workshop, South Korea.
- 2018 ■ **Research Award (for the 8th Bar-Ilan Winter School on Cryptography)**, by Koç University.
- **TUBITAK Research-Assistantship Stipend** (for M.Sc.).
- 2017 ■ **Graduate School of Science and Engineering (GSSE) Outstanding Teaching Assistant Award**, by Koç University.
- **Travel Grant**, ESORICS, Data Privacy Management Workshop.
- 2015 ■ **The Valedictorian of Computer Science and Engineering Department**, Gaziantep Zirve University.
- 2014 ■ **Mitacs Globalink Summer Research Internship Award**, Canada.

References

Prof. Nikolaos Laoutaris (Ph.D. Advisor), IMDEA Networks Institute.

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Prof. Gene Tsudik, University of California, Irvine.

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Prof. Alptekin Küpçü, Koç University.

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Prof. Zekeriya Erkin, TU Delft.

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